

STA2020 Tutorial 2: Multiple Linear Regression

Box and Youle (1955) performed a series of 19 experiments to investigate a chemical reaction. The aim was to establish if a relationship exists between the percentage of material converted to the desired substance (y) and the temperature of the reaction (in °C), the time of the reaction (in minutes), and the concentration of the element in question (in %). The results are shown in the following table:

Experiment	%Converted (Y)	Temperature (X_1)	Concentration (X_2)	Time (X_3)
1	45.9	162	23	3
2	53.3	162	23	8
3	57.5	162	30	5
4	58.8	162	30	8
5	60.6	172	25	5
6	58	172	25	8
7	58.6	172	30	5
8	52.4	172	30	8
9	56.9	167	27.5	6.5
10	55.4	177	27.5	6.5
11	46.9	157	27.5	6.5
12	57.3	167	32.5	6.5
13	55	167	22.5	6.5
14	58.9	167	27.5	9.5
15	50.3	167	27.5	3.5
16	61.1	177	20	6.5
17	62.9	177	20	6.5
18	60	160	34	7.5
19	60.6	160	34	7.5

The correlations between all the variables are as follows:

	Converted	Temperature	Concentration	Time
Converted	1.000	0.404	0.080	0.393
Temperature	0.404	1.000	-0.462	-0.022
Concentration	0.080	-0.462	1.000	0.177
Time	0.393	-0.022	0.177	1.000

A multiple linear regression was performed on this dataset, yielding the following results:

```
Call:
lm(formula = Converted ~ Temperature + Concentration + Time,
    data = MLR)

Residuals:
    Min       1Q   Median       3Q      Max
-8.2075 -2.0557  0.1633  3.2712  4.7503

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)  -26.0353    32.9734  -0.790   0.4421
Temperature    0.4046     0.1747   2.316   0.0351 *
Concentration  0.2930     0.2604   1.125   0.2783
Time          1.0338     0.5994   1.725   0.1051
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 4.078 on 15 degrees of freedom
Multiple R-squared:  A,    Adjusted R-squared:  B
F-statistic: 3.027 on 3 and 15 DF,  p-value: 0.06235
```

ANOVA					
	<i>df</i>	<i>SS</i>	<i>MS</i>	<i>F</i>	<i>Significance F</i>
Regression	3	151.002	50.334	3.027	0.062
Residual	15	249.462	16.631		
Total	18	400.464			

Use the above provided information to answer the following questions:

- 1) Give the estimated regression equation.

$$\hat{y} = -26.0353 + 0.4046(\text{Temperature}) + 0.2930(\text{Concentration}) + 1.0338(\text{Time})$$

- 2) According to this model, what will the predicted percentage of converted material be in an experiment where a concentration of 28% is run for 6 minutes at 170°C?

$$\hat{y} = -26.0353 + 0.4046(170) + 0.2930(28) + 1.0338(6) = 57.1535$$

∴ The predicted percentage of converted material will be approximately 57.15%.

- 3) According to the model, how much longer does the experiment need to be run in order to increase the percentage of converted material by 25%? Evaluate the feasibility of this answer.

If concentration and temperature are held constant, a one-minute increase in time would lead to a 1.0338 percentage increase in material converted to the desired substance (y), on average.

$$\therefore 1.0338(\text{Time}) = 25$$

$$\therefore \text{Time} = 24.18263 \text{ minutes}$$

This value of Time is far beyond the range of data that has been observed (check the table of results). Hence, extrapolating to a value this high is not advisable.

Additionally, assuming a linear trend for the relationship between Time and y beyond the available data is not feasible. The value of y cannot be larger than 100 (% measurement), therefore the relationship between Time and y will reach some asymptote as Time increases.

- 4) Is there a sensible interpretation of the estimate for the intercept coefficient?

No. The estimated intercept coefficient refers to the value of y when all the (numeric) independent variables are equal to zero. This is intuitively not possible for Time (if time is zero, then no experiment was run) and Concentration (a concentration of zero percentage means there is no material of interest).

- 5) Calculate the missing values in the R output (specifically the value of A and B). Additionally, based on this, provide an estimate of the sample correlation coefficient.

For the value of A:

$$R^2 = \frac{SS_{\text{regression}}}{SS_{\text{total}}} = \frac{151.002}{400.464} = 0.377$$

For the value of B:

$$\begin{aligned} R_a^2 &= 1 - \left(\frac{n-1}{n-p-1} \right) (1 - R^2) \\ &= 1 - \left(\frac{19-1}{19-3-1} \right) (1 - 0.377) = 0.252 \end{aligned}$$

For the value of estimated sample correlation coefficient:

$$r = \sqrt{R^2} = \sqrt{0.377} = 0.614$$

- 6) Comment on the overall significance of this fitted regression model by referring to the appropriate values.

When testing for the overall significance of the model – use the F-test!

$$H_0: \beta_1 = \beta_2 = \beta_3 = 0$$

$$H_a: \text{At least one } \beta_j \neq 0$$

Test statistic (from R output) is $F = 3.027 \sim F_{3,15}$ which is associated with a p-value of 0.06235.

Although this regression model is not strictly significant at an assumed 5% level of significance ($\alpha = 0.05$), it is very close to this cut-off value.. Thus, it is perhaps worth investigating some alterations to the model to try and improve the fit such that there is more evidence against the null hypothesis and so can conclude that this model is overall significant.

Additionally, the coefficient of determination ($R^2 = 0.377$) is quite low, suggesting that the inclusion of all three independent variables do not account (or explain) for much of the variation in y .

- 7) Which of the independent variables in the fitted model are significant at $\alpha = 0.05$?

When testing for the significance of individual, independent variables – use t-tests!

$$H_0: \beta_j = 0$$

$$H_a: \beta_j \neq 0$$

Technically repeating this hypothesis test three times, as there are three independent variables. From the R output, the following can be summarized related to these three hypothesis tests:

Variable	Test statistic ($\sim t_{15}$)	p-value	Significant? ($\alpha = 0.05$)
<i>Temperature</i>	2.316	0.035	Yes
<i>Concentration</i>	1.125	0.278	No
<i>Time</i>	1.725	0.105	No

- 8) Is there any danger of multicollinearity in this analysis? Justify your answer by referring to the relevant values.

When looking at the correlations between the independent variables, it can be seen that there is a negative, moderate correlation ($r = -0.462$) between *Temperature* and *Concentration*, which may be a cause of concern. The inclusion of rather highly correlated independent variables means that there is redundant

information contained within the model. This in turn adversely affects the regression results.

9) How would you go about trying to improve the model fit?

- Remove one or more of the less significant variables from the model.
- Remove a redundant variable from the model.
- Consider using other, non-linear forms of regression.
- Consider adding interaction terms.

As an extra exercise, use the R script (containing the dataset) and perform another multiple regression analysis – but excluding the least significant variable. Compare this new model to the results given above.

From the previous R output, it is apparent that the *Concentration* variable is associated with the least significant test-statistic (related to a very high p-value). Additionally, it has a very weak correlation coefficient value with the outcome of interest. Hence, re-fit the MLR model in R using only *Temperature* and *Time* (i.e., omit *Concentration*) as the independent variables and find the following output:

```
> fit <- lm(Converted ~ Temperature + Time, data = MLR)
> summary(fit)

Call:
lm(formula = Converted ~ Temperature + Time, data = MLR)

Residuals:
    Min       1Q   Median       3Q      Max
-7.144 -2.785  0.662  3.034  5.393

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)  -3.5856    26.4657  -0.135   0.8939
Temperature    0.3131     0.1559   2.008   0.0618 .
Time          1.1605     0.5936   1.955   0.0683 .
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 4.112 on 16 degrees of freedom
Multiple R-squared:  0.3245,    Adjusted R-squared:  0.2401
F-statistic: 3.843 on 2 and 16 DF,  p-value: 0.04334
```

The overall significance of the model has improved (p-value = 0.043) and is now significant at $\alpha = 0.05$ (compare to results obtained in Question 6). Although neither *Temperature* nor *Time* are strictly significant at $\alpha = 0.05$, there is still relatively strong evidence to suggest that there is a linear relationship between these said variables and the outcome of interest.